Nishant Choudhary

B.Tech Electronics & Computer Engineering | VIT Chennai | 2027

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EDUCATION

B.Tech – Electronics & Computer Engineering, VIT Chennai — Aug 2023 – May 2027

CGPA: **8.32 / 10** Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

TECHNICAL SKILLS

Programming Languages	Python, C++, SQL
AI / Machine Learning	Machine Learning, Deep Learning, Computer Vision, NLP, Transformers, Transfer Learning
Frameworks & Tools	PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, FastAPI, Streamlit, Plotly
Software Engineering	OOP, Data Structures & Algorithms, REST APIs, Git/GitHub, Docker, Linux (CLI), pytest
Core Computer Science	Operating Systems, Computer Networks, DBMS, Database Design
Databases	MySQL, SQL (Schema Design, Query Optimization, Indexing, ETL)

SELECTED PROJECTS

ClickTutor AI – AI-Powered Desktop Learning Assistant

Python · PyQt6 · Gemini Vision API · PyTesseract · OpenCV — GitHub

- Architected a modular **MVC-based** desktop app integrating OCR, multimodal LLM inference, and real-time overlay rendering for context-aware explanations.
- Engineered a resilient **multi-pass OCR localization pipeline** (exact, fuzzy, substring, and duplicate-aware matching), achieving **100% accuracy (116/116)** across **30 benchmark iterations**.
- Profiled and optimized the full pipeline; overlay rendering averaged **<10 ms**, isolating OCR and LLM inference as the primary latency contributors (**∼800 ms** total) to guide optimization.
- Built a **pytest** test suite covering exact, fuzzy, and duplicate-resolution cases across **116 test samples**.

FocusLens – Real-Time Attention Monitoring System

Python · MediaPipe · Random Forest · LSTM · Streamlit — GitHub

- Built a real-time **computer vision pipeline** using MediaPipe Face Mesh, extracting **478 facial landmarks** per frame at **54.17 FPS** face-detection throughput with zero dropped frames.
- Engineered a feature-extraction layer (EAR, PERCLOS, blink rate, head-pose) at **2.65 ms** latency, feeding a Random Forest + LSTM inference stage with **78.28 ms** end-to-end prediction latency.
- Achieved **97.67% accuracy**, **98.82% F1**, and **100% recall** on held-out evaluation data, with **96% tracking stability** across live video streams.

Fake News Detection REST API

Python · FastAPI · BERT · Hugging Face · Docker · REST API — GitHub

- Designed and deployed a **FastAPI** REST service for a fine-tuned **BERT** 6-class claim classifier (LIAR dataset) using **Hugging Face Transformers**, with a modular pipeline for preprocessing, training, evaluation, and serving.
- Containerized with **Docker**; benchmarked serving at **8.5 ms** average / **12.8 ms** P95 latency and **396 req/sec** batch throughput, with a **0.67 s** model load and **418 MB** footprint.

Government Budget Analytics Platform

Python · MySQL · SQL · Streamlit · ETL · Plotly — GitHub

- Designed a normalized **6-table MySQL schema** with foreign keys, SQL views, and indexed relationships for multi-year Government of India tax revenue data.
- Built a transaction-safe **ETL pipeline** with rollback recovery and idempotent upserts; optimized analytical SQL queries at **1M-row** scale, cutting JOIN latency by **∼83%** (**2066 ms → 350 ms**).
- Built a **Streamlit** dashboard with KPI visuals and a SQL query explorer, cutting response time by **∼91%** (**420 ms → 38 ms**).

ACHIEVEMENTS & CERTIFICATIONS

- Solved **150+** Data Structures and Algorithms problems on LeetCode.
- Built **4 end-to-end software projects** spanning desktop applications, database systems, REST APIs, and computer vision/NLP systems.
- Earned the **Oracle OCI AI Foundations Associate** certification (2025).
- Completed Coursera's **Machine Learning Specialization** (Andrew Ng) and **Git & GitHub** Certificate.